MIL-R-14593A
24 June 1965
SUPERSEDING
MIL-W-14593(ORD)
13 August 1957

MILITARY SPECIFICATION

RESISTANCE WIRE (ENAMELED)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers resistance wire of an alloy of nickel, chromium, and silicon, insulated with oleo-resinous enamel and having varying degrees of uniformity with respect to resistance.

1.2 Classification.- The nickel-chromium-silicon resistance wire shall be of the following grades, as specified (see 6.2).

Grade A - 0.05000 ± 0.00010 (ratio) uniformity of resistance.
Grade B - 0.05000 ± 0.00025 (ratio) uniformity of resistance.
Grade C - No specified uniformity of resistance.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-12000	Cable, Cord, and Wire, Electric Packaging and Packing of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 6145

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

B-63-49 Method of Test for Resistivity of Metallically Conducting Resistance and Contact Materials

(Copies of ASTM Specifications may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

3. REQUIREMENTS

3.1 Chemical requirements.- The chemical composition of the bare resistance wire shall be in accordance with Table I.

TABLE I - Chemical Composition

Element	Percent	
	Minimum	Maximum
Nickel 1/	77	79
Chromium	19	20
Silicon	0.75	1.5
Iron	--	1.0
Manganese	--	2.5
Carbon	--	0.25
Sulfur	--	0.03

1/ Includes small amounts of cobalt

3.2 Dimensional tolerances.- The enameled resistance wire shall meet the following dimensional requirements in accordance with Table II. (See 6.2).

3.3 Loop strength.- The loop strength of the enameled wire shall be in accordance with Table II.

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TABLE II - Performance Criteria

Diam of Bare Wire Inch (Nom.)	Max Diam of Wire After Enameling (Inch)	Loop Strength Pounds (Min.)	Electrical Resistance Ohms/Ft. at 25°C		Mandrel Diameters Inch (Max.)
			Min.	Max.	
0.002	0.0025	0.35	159.2	175.5	0.015
0.0025	0.0030	0.60	101.9	112.3	0.015
0.0031	0.0038	1.10	66.2	73.0	0.015
0.004	0.0048	1.80	39.8	43.9	0.015
0.005	0.0058	2.00	25.5	28.1	0.015
0.008	0.0088	4.00	9.96	10.97	0.025

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3.4 Electrical requirements.

3.4.1 Electrical resistance.- The electrical resistance of the enameled wire shall be in accordance with Table II.

3.4.1.1 Coefficient of electrical resistance.- The coefficient of resistance per degree centigrade of a three foot specimen in the range of 20 to 100°C shall not exceed a maximum 0.000125 change of resistance.

3.4.1.2 Uniformity of electrical resistance.- The resistance value of each 50 ft. of every 1000 ft. of the unenameled wire when measured in increments of exactly 1/20th, 2/20ths and so forth shall not differ from its proportion of total resistance by more than the value specified in Table III.

TABLE III - Uniformity of Resistance

Grade	Ratio	Tolerance
A	0.05000	± 0.00010
B	0.05000	± 0.00025
C	--	--

3.5 Flexibility.- The enameled resistance wire shall be capable of being wound on a mandrel of the diameter specified in Table II without showing visible cracks in the wire or the enamel, or any evidence of peeling of the enamel coat, when exposed to a temperature of 20 to 30°C.

3.6 Chemical resistance of enamel coat.- The enamel coat shall be insoluble in hot transformer oil at a minimum temperature of 100°C when exposed for periods of 48 hours or more.

3.7 Workmanship.- The unenameled conductor shall be bright and free from oil, circular in cross-section, uniform in diameter and composition, and free from scale, pits, slivers, corrosion, die marks, and other imperfections. Joints will not be allowed.

3.7.1 The conductor shall be coated with a continuous, smooth film of enamel in a thoroughly workmanlike manner. The coating shall be uniform, smooth, and free from any bubbles or lumps of enamel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for

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the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Lot acceptance inspection.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, the inspection lot shall be determined in accordance with "Submission of Product" as specified in MIL-STD-105 except that the unit of product for determining lot size shall be 1 spool or 1 reel. The sample unit therefrom shall be a length of wire sufficient for performing all tests.

4.2.2 Examination and tests.- Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-C-12000 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE IV - Classification of Defects

CRITICAL: NONE

MAJOR: AQL 1.0% Inspection Level II	REQUIREMENT	TEST PROCEDURE
101. Dimensional tolerances	3.2	4.4.2
102. Loop strength	3.3	4.4.3
103. Electrical resistance	3.4.1	4.4.4
104. Flexibility	3.5	4.4.5
105. Workmanship	3.7	4.4.7

MINOR: NONE

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4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Process control tests.- One spool or reel shall be selected at random as a control sample from each 500 produced, or from each months production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST		REQUIREMENT	TEST PROCEDURE
301.	Chemical requirements	3.1	4.4.1
302.	Coefficient of electrical resistance	3.4.1.1	4.4.4.1
303.	Uniformity of electrical resistance	3.4.1.2	4.4.4.2
304.	Chemical resistance of enamel coat	3.6	4.4.6

4.3.1 Control test failure.- Should any one control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made to all wire represented by the sample, and resubmitted samples therefrom have been approved.

4.4 Test methods and procedures.

4.4.1 Chemical requirements.- Laboratory tests shall be made to determine the composition by approved analytical methods. Composition shall be in accordance with Table I. Laboratory designation shall be as specified in 6.2.

4.4.2 Dimensional tolerances.- The dimensional tolerances as specified in 3.2 shall be measured using standard measuring equipment that is accurate to within .0001 inch. Measurements shall be taken at least 1 yard apart for 3 yards from each sample spool. Two readings shall be taken at each measurement at least 90 degrees apart at the same location on the diameter. Readings shall not be averaged out.

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4.4.3 Loop strength.- Loop strength as specified in 3.3 shall be determined using standard measuring equipment. Cut two 15 inch lengths of wire from a sample spool which shall be free of bends, kinks or other imperfections. Fasten both ends of one length of wire in the upper jaw of the testing machine to form a loop approximately 6 inches long. Pass one end of the second length of wire through the loop and fasten the free ends in the lower jaw. Apply the load at a uniform rate so that the break occurs in not less than 20 seconds. The break should be at the point where the loops contact. At least five tests shall be made to determine the minimum breaking load. Measurements shall not be averaged. Both lengths of wire shall be discarded after each test.

4.4.4 Electrical resistance.- Using standard measuring equipment, the electrical resistance of the wire shall be measured in accordance with ASTM B-63-49. The wire need not be weighed to determine its electrical resistivity. Resistance shall be in accordance with 3.4.1.

4.4.4.1 Coefficient of resistance.- Measurements shall be made on each specimen using standard measuring equipment, capable of measuring the change in resistance to an accuracy of ± 0.01 percent of the resistance of the specimen. Wind the specimen from each sample loosely, in a single layer on a form made of insulating material such as phenol fiber, hard rubber, mica or glass, etc. The wound specimen shall not be under tension. Immerse the specimen in a bath of oil at the low temperature (T1) specified in 3.4.1.1 until the specimen is at thermal equilibrium. Measure the resistance at this temperature (R1). Heat the bath or have a second bath maintained at the high temperature specified in 3.4.1.1 (T2). Allow the specimen to remain immersed until it is at thermal equilibrium. Measure resistance (R2) at this temperature. T1 and T2 shall be any two temperatures in the range of 20 to 100°C inclusive, such that the spread between them shall be at least 70°C. The temperature T1 and T2 shall be maintained and determined to within an accuracy of ± 0.5°C. The coefficient of resistance shall be in accordance with 3.4.1.1 and Table II as determined by the following formula:

$$\text{Coefficient of resistant per degree C} = \frac{R2 - R1}{R1\,(T2 - T1)}$$

4.4.4.2 Uniformity of electrical resistance.- This test shall be performed with standard measuring equipment having a precision of ± 0.002 percent. Measure the resistance of a length of wire 1000 feet long, measuring each section of the wire in increments of 1/20th, 2/20ths, and so forth. The resistance of each section shall not differ from its proportion of the total resistance by more than the values specified in 3.4.1.2.

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4.4.5 Flexibility.- Wind approximately one foot of finished wire in an even and compact winding of not less than 20 turns around a mandrel of the size specified in Table II. The wire shall then be visually examined for cracks, chipping, or peeling of the enamel coating. This test shall be performed between the temperatures specified in 3.5.

4.4.6 Chemical resistance of enamel coat.- Immerse a length of wire from each sample, approximately 12 inches long in hot oil for the time duration and at the temperature specified in 3.6. Within 2 minutes after removing the wire from the hot oil draw the wire, without stretching, between a section of cheese cloth consisting of 4 thicknesses folded over the wire which is pressed firmly between the thumb and forefinger. The wire shall be visually examined to insure that no enamel has been removed to the extent of exposing the bare conductor in compliance with 3.6. The wires shall then be tested in accordance with 4.4.4 to determine conformance with 3.4.1.

4.4.7 Workmanship.- The wire shall be inspected by visual and tactile means to determine conformance with the workmanship requirements specified in 3.7 and 3.7.1.

5 PREPARATION FOR DELIVERY

5.1 Levels A, B, and C.

5.1.1 Packaging and packing.- The resistance wire shall be cleaned, preserved, packaged, packed and marked in accordance with MIL-C-12000.

6. NOTES

6.1 Intended use.- The enameled resistance wire is intended for resistance windings in precision variable resistors and where the smallest change in resistance due to the strain of winding is desired.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade and quantity of resistance wire required.
(c) Dimensional requirements.
(d) Selection of applicable levels of preservation, packaging, and packing.
(e) Laboratory for testing chemical requirements.

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Custodians:
Army - MU
Navy - WP
Air Force - 85

Review activities:
Army - MU, MI, EL
Navy - WP
Air Force - 85

User activities:
Army - MO
Navy - MC
Air Force - None

Preparing activity:
Army - MU

Project 6145-0281

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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